Video Condition

Agenda:

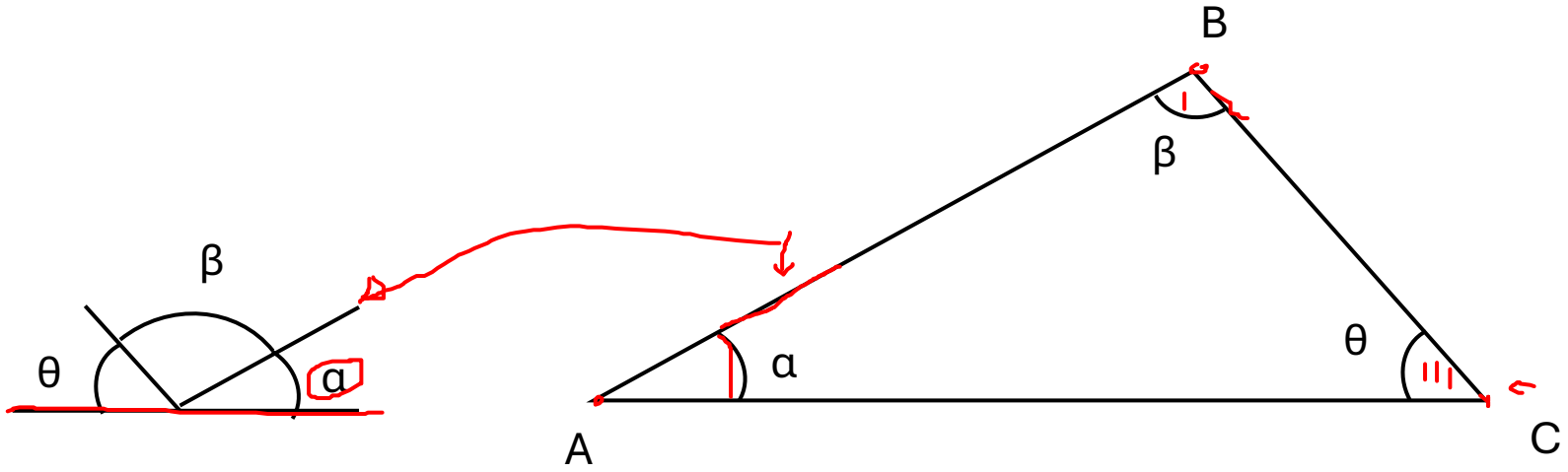
1. Theorem # 1 Internal Δ angles sum
2. Theorem # 2 External Δ angles sum
3. Theorem # 3 Exterior angle
4. Classification of triangles based on their sides

Bibliography: Trucios, C., & Trujillo, M. (2021). Chapter 2/ Triangle. In *Plane and spatial geometry* (7th ed., pp. 34–56). essay, Fondo Editorial Rodo.

Theorem # 1 Internal Δ angles sum

In every triangle, the sum of the measures of the angles is equal to 180°

In the diagram:



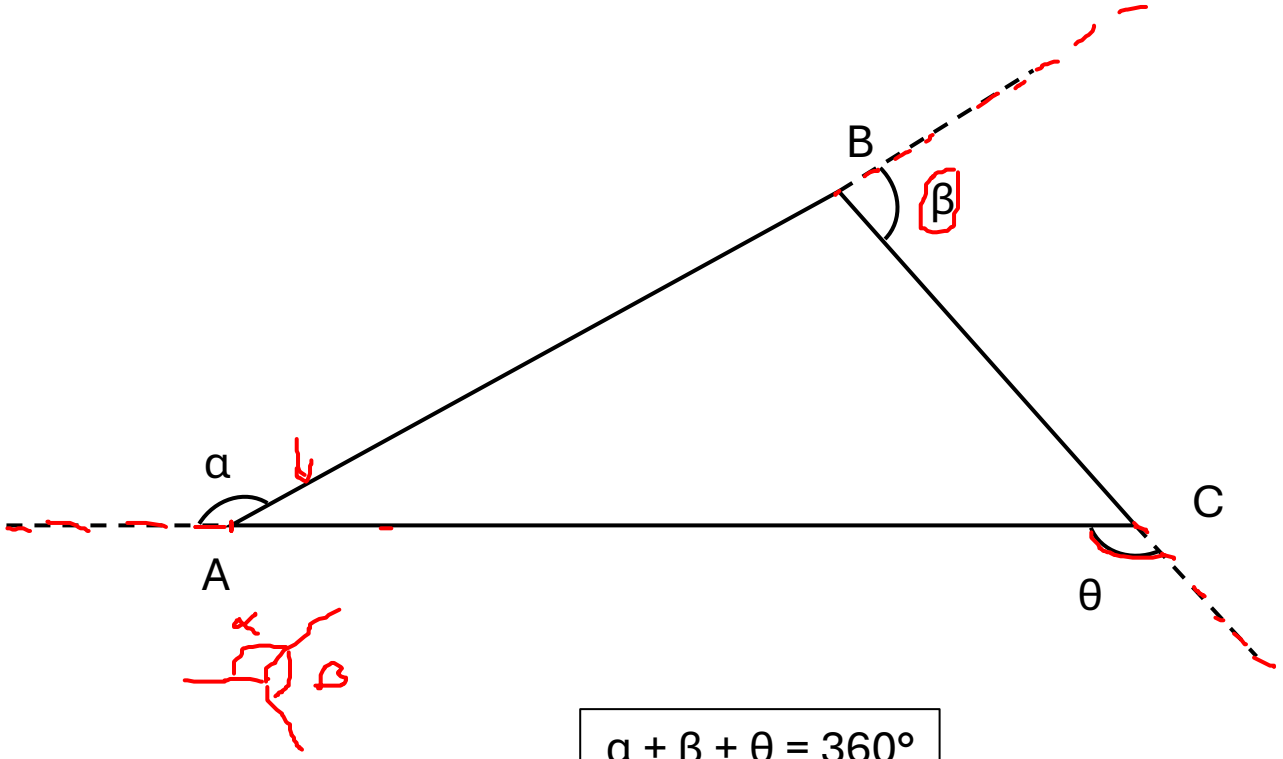
$\theta + \alpha + \beta = 180°$

$\theta + \alpha + \beta = 180°$

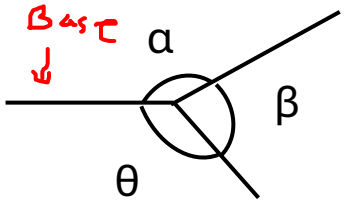
Theorem # 2 External Δ angles sum

In any triangle, the sum of the exterior angles, one at each vertex, is 360°

In the diagram:



$$\alpha + \beta + \theta = 360°$$

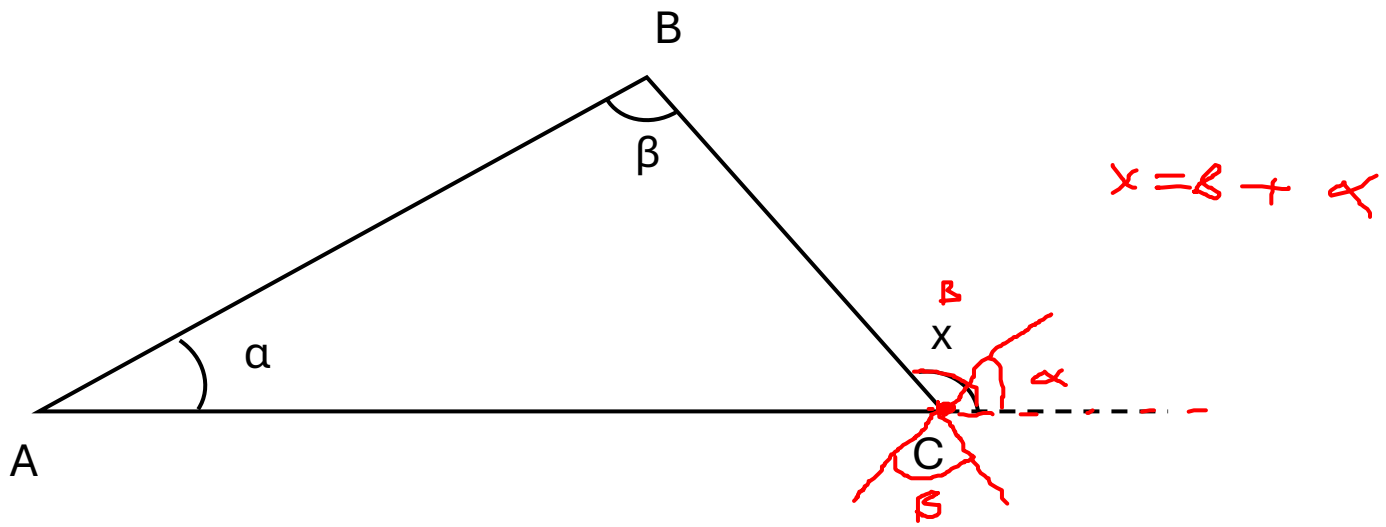


$$\alpha + \beta + \theta = 360°$$

Theorem # 3 Exterior angle

In every triangle, the measure of an exterior angle is equal to the sum of the measures of the two interior angles that are not adjacent to that exterior angle.

In the diagram:

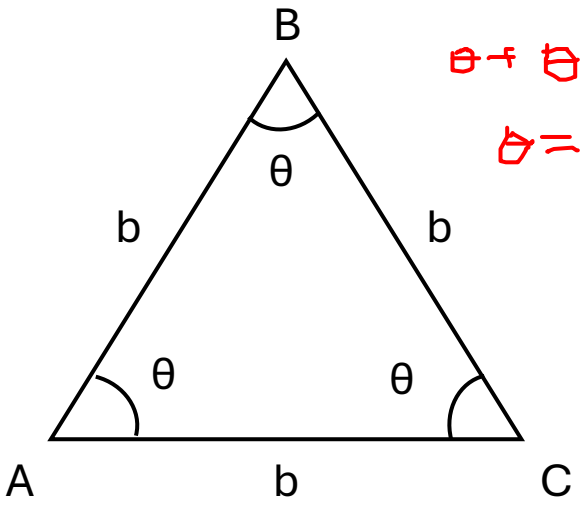


$$\boxed{x = \alpha + \beta}$$

Classification of triangles based on their sides

Equilateral triangle

It has three sides of equal length.



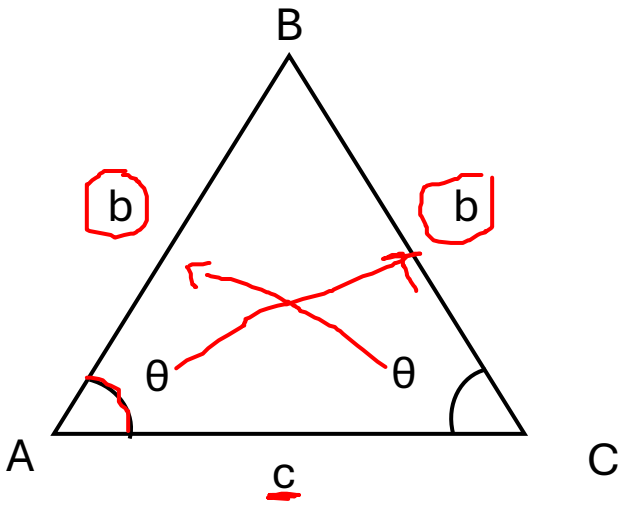
$\theta + \theta + \theta = 180°$

$\theta = 60°$

$\theta = 60°$

Isosceles triangle

Two of its sides, and only two, have equal lengths.



Where: $\overline{AB}$ y $\overline{BC}$: are the sides
$\overline{AC}$: is the base

$b \neq c$

$\theta < 90°$

Thanks for watching!
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Video Link: https://drive.google.com/file/d/1yjINGQ0PZpxNfBV-8lg8ZiR3-NfIMe4B/view?usp=drive_link